Chapter 4: Expressions Using Letter – Numbers

Multiple Choice Questions

1. **What is the expression for a number increased by 7?**

   A) $x - 7$
   B) $x + 7$
   C) $7x$
   D) $x \div 7$

   Answer: B) $x + 7$

   Explanation: Adding 7 to a number x is represented algebraically as $x + 7$.

2. **Which expression represents 3 less than a number y?**

   A) $3y$
   B) $y + 3$
   C) $y - 3$
   D) $3 - y$

   Answer: C) $y - 3$

   Explanation: "3 less than y" means we subtract 3 from y, giving $y - 3$.

3. **If m represents a number, which expression shows 5 more than double the number?**

   A) $5m + 2$
   B) $2m + 5$
   C) $m + 5 + m$
   D) $2 + 5m$

   Answer: B) $2m + 5$

   Explanation: Doubling m gives $2m$. Adding 5 more gives the expression $2m+5$.

4. **Which of the following best expresses "a number t multiplied by 6"?**

   A) $6 + t$
   B) $6t$
   C) $t \div 6$

D) $t - 6$

Answer: B) $6t$

Explanation: The multiplication of 6 and t is written as $6t$.

5. **Which of these expressions represents the total cost of buying x pencils if each pencil costs ₹4?**

A) $4 + x$
B) $x \div 4$
C) $4x$
D) $x - 4$

Answer: C) $4x$

Explanation: If one pencil costs ₹4, then x pencils cost ₹$4x$ in total.

6. **Aftab is a years old. Shabnam is 3 years older than Aftab. What is Shabnam's age in terms of a?**

A) $a - 3$
B) $3a$
C) $a + 3$
D) $a \div 3$

Answer: C) $a + 3$

Explanation: Shabnam is 3 years older, so we add 3 to Aftab's age: $a + 3$.

7. **If the cost of a coconut is ₹35 and c coconuts are bought, what is the total cost?**

A) $35 + c$
B) $35c$
C) $c \div 35$
D) $c - 35$

Answer: B) $35c$

Explanation: Cost of one coconut is ₹35, so for c coconuts, the total is $35 \times c = 35c$.

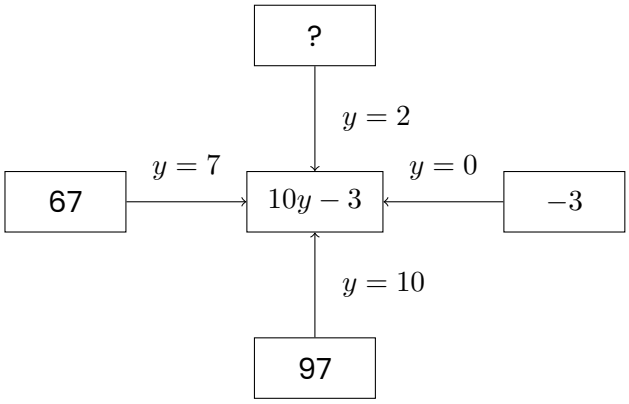
8. Jaggery costs ₹60 per kg. If j kg is purchased, which expression gives the cost?

A) $60j$
B) $j + 60$
C) $60 \div j$
D) $j - 60$

Answer: A) $60j$

Explanation: ₹60 per kg means we multiply 60 by the number of kg: $60j$.

9. In the diagram, each input passes through the rule $10y - 3$. What number should go in the missing box?



A) 17
B) 20
C) 0
D) 7

Answer: A) 17

Explanation: Using the rule $10y - 3$, for $y = 2$:

$$10 \cdot 2 - 3 = 20 - 3 = 17$$

So, the missing value is 17

10. A shop rents a chair for ₹40 and refunds ₹6 on return. What is the net cost for renting x chairs?

A) $46x$

B) $34x$

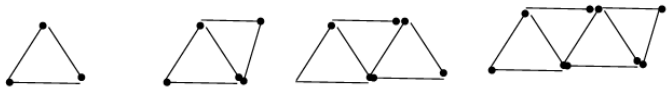
C) $40x + 6x$

D) $40 - 6x$

Answer: B) $34x$

Explanation: Net cost per chair = ₹40 - ₹6 = ₹34. So for x chairs: $34x$.

11. Observe the pattern of matchsticks below. What is the expression for the number of matchsticks in Step n?



A) $n + 2$

B) $3n$

C) $2n + 1$

D) $n \times n$

Answer: C) $2n + 1$

Explanation: The pattern starts with 3 matchsticks and adds 2 for every next triangle. General rule: $2n + 1$.

12. Each "L" shape in the figure below is made using 2 matchsticks. What is the total number of matchsticks for n L-shapes?



A) $n + 2$

B) $2n$

C) $n \div 2$

D) n^2

Answer: B) $2n$

Explanation: Each L uses 2 matchsticks, so total matchsticks = $2 \times n = 2n$.

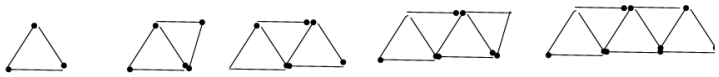
13. In a matchstick pattern, the number of matchsticks follows the rule $3+2(n-1)$. What is the expression in its simplified form?

A) $2n+1$
B) $3n-2$
C) $2n-1$
D) $3n$

Answer: A) $2n+1$

Explanation: Simplify: $3+2(n-1)=3+2n-2=2n+1$.

14. Refer to the matchstick pattern shown. How many matchsticks will be used in Step 5?

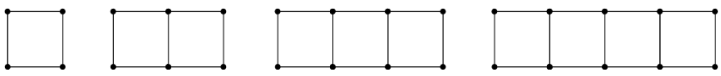


A) 9
B) 11
C) 13
D) 15

Answer: B) 11

Explanation: Using the formula $2n+1$: $2(5)+1=11$.

15. A pattern of squares is created with shared sides. The number of matchsticks used in w squares is given in the image below. What is the correct expression?



A) $4w$
B) $3w+1$
C) $2w+2$
D) $5w-1$

Answer: B) $3w + 1$

Explanation: First square uses 4 matchsticks; each additional square shares 1 side, so adds only 3 sticks: $3w + 1$.

16. What is the expression for the perimeter of a square with side length s?

A) $2s$
B) s^2
C) $4s$
D) $s + 4$

Answer: C) $4s$

Explanation: A square has 4 equal sides. Perimeter = $s + s + s + s = 4s$.

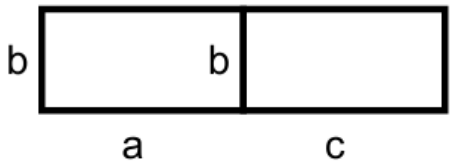
17. What is the area of a rectangle with length l and breadth b?

A) $l + b$
B) $2l + 2b$
C) lb
D) $l^2 + b^2$

Answer: C) lb

Explanation: Area of a rectangle = length × breadth = lb.

18. Refer to the diagram of a rectangle split into two smaller rectangles. What is the expression for the total area?



A) $ab + bc$
B) $ab + ac$
C) $ab + ad$
D) $ab + cd$

Answer: A) $ab + bc$

Explanation: Each sub-rectangle has area = length × width. Total area is the sum of both: $ab + bc$.

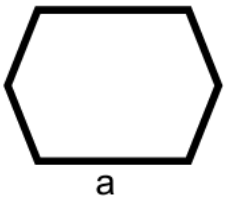
19. A rectangle has length l and breadth b. What is its perimeter?

A) $2l + 2b$
B) lb
C) $4l$
D) $l + b$

Answer: A) $2l + 2b$

Explanation: Perimeter of a rectangle is sum of all sides: $l + b + l + b = 2l + 2b$.

20. A regular hexagon has side length a. What is the expression for its perimeter?



A) $5a$
B) $4a$
C) $6a$
D) a^2

Answer: C) $6a$

Explanation: A regular hexagon has 6 equal sides. Perimeter = $6a$.

21. Simplify the expression: $5c + 3c + 10c$

A) 18
B) $15c$
C) $18c$
D) $50c$

Answer: C) $18c$

Explanation: All terms have the same variable. Add coefficients: $5 + 3 + 10 = 18$. So, $18c$.

22. Simplify the expression: $4x + 3y - 2x + y$

A) $2x + 4y$
B) $6x + 4y$
C) $2x + 2y$
D) $7x + y$

Answer: A) $2x + 4y$

Explanation: Group like terms: $4x - 2x = 2x$, $3y + y = 4y$. So, simplified form is $2x + 4y$.

23. Which of the following is the simplified form of $12n - 4n$**?**

A) $8n$
B) $16n$
C) $4n$
D) $-8n$

Answer: A) $8n$

Explanation: Subtract coefficients: $12n - 4n = 8n$.

24. What is the simplified form of $7p - 3p + 2q + q$**?**

A) $4p + 3q$
B) $10pq$
C) $4p + q$
D) $4p + 2q$

Answer: A) $4p + 3q$

Explanation: Combine like terms: $7p - 3p = 4p$, $2q + q = 3q$.

25. Simplify: $2x + 3y - x + 5y$

A) $x + 2y$
B) $3x + 8y$

C) $x + 8y$

D) $x + 5y$

Answer: C) $x + 8y$

Explanation: $2x - x = x$, $3y + 5y = 8y$. So the simplified form is $x + 8y$.

26. If $a = 5$, what is the value of the expression $3a + 4$?

A) 15

B) 19

C) 17

D) 14

Answer: B) 19

Explanation: Substitute $a = 5$: $3 \times 5 + 4 = 15 + 4 = 19$.

27. Evaluate the expression $2m - 7$ for $m = 4$.

A) 8

B) 0

C) 1

D) -5

Answer: C) 1

Explanation: $2 \times 4 - 7 = 8 - 7 = 1$.

28. Find the value of $4x + 2y$ when $x = 2$ and $y = 3$.

A) 12

B) 14

C) 10

D) 16

Answer: B) 14

Explanation: $4(2) + 2(3) = 8 + 6 = 14$.

29. If $p = 6$, what is the value of $5p - 2p$?

A) 18

B) 30
C) 20
D) 24

Answer: A) 18

Explanation: $5p - 2p = 3p = 3 \times 6 = 18$.

30. What is the value of $3a - 2b$ if $a = 4$ and $b = 5$?

A) 2
B) 3
C) 4
D) 7

Answer: A) 2

Explanation: $3(4) - 2(5) = 12 - 10 = 2$.

31. Which of the following sets contains only like terms?

A) $4x, 5x, 7y$
B) $3x, 2x, 6x$
C) x^2, x, x^3
D) $4a, 4b, 4c$

Answer: B) $3x, 2x, 6x$

Explanation: All terms have the same variable and power: x. These are like terms.

32. Which pair of terms are unlike terms?

A) $5m, 7m$
B) $2a, 3a$
C) $4xy, 6yx$
D) $2x, 5x^2$

Answer: D) $2x, 5x^2$

Explanation: The powers of the variable are different. So, x and x^2 are unlike.

33. What are like terms in the expression $3a + 4b + 7a$?

A) $3a, 4b$
B) $3a, 7a$
C) $4b, 7a$
D) $3a, 4b, 7a$

Answer: B) $3a, 7a$

Explanation: Like terms must have the same variable and power. $3a$ and $7a$ are like terms.

34. Which option correctly groups the like terms from $6p + 2q + 4p + 5q$?

A) $(6p + 2q), (4p + 5q)$
B) $(6p + 4p), (2q + 5q)$
C) $(6p + 5q), (4p + 2q)$
D) None of these

Answer: B) $(6p + 4p), (2q + 5q)$

Explanation: Grouping like terms gives $6p + 4p$ and $2q + 5q$.

35. What is the result of adding the like terms $8r + 3r$?

A) $11r$
B) $24r$
C) $5r$
D) r

Answer: A) $11r$

Explanation: Like terms add by combining coefficients: $8r + 3r = 11r$.

36. Simplify the expression: $3(x + 4)$

A) $3x + 4$
B) $3x + 12$
C) $3 + 4x$
D) $x + 12$

Answer: B) $3x + 12$

Explanation: Apply distributive property: $3 \times x + 3 \times 4 = 3x + 12$.

37. What is the result of removing brackets from: $2a + (3b + 4c)$**?**

A) $2a + 3b + 4c$
B) $5a + 7b + 4c$
C) $2a + 7b$
D) $2a + 3b + 7c$

Answer: A) $2a + 3b + 4c$

Explanation: There is no minus or coefficient outside the bracket, so remove it directly.

38. Simplify: $5(x - 2)$

A) $5x - 10$
B) $5x + 10$
C) $x - 10$
D) $5x - 2$

Answer: A) $5x - 10$

Explanation: Apply distributive property: $5 \times x - 5 \times 2 = 5x - 10$.

39. What is the simplified form of: $4m - (2m + 5)$**?**

A) $2m + 5$
B) $6m - 5$
C) $2m - 5$
D) $2m + 3$

Answer: C) $2m - 5$

Explanation: Remove brackets carefully: $4m - 2m - 5 = 2m - 5$.

40. Simplify the expression: $6(p + 3) - 2p$

A) $4p + 18$
B) $6p + 18$

C) $6p + 3 - 2p$

D) $4p + 3$

Answer: A) $4p + 18$

Explanation: First apply distributive property: $6p + 18 - 2p = 4p + 18$.

41. Look at the expressions below. Are $5u$ and $5 + u$ equal?

A) Yes, both represent the same quantity.

B) No, $5u$ is multiplication, $5 + u$ is addition.

C) Yes, if $u = 1$

D) They are equal for all values of u

Answer: B) No, $5u$ is multiplication, $5 + u$ is addition.

Explanation: $5u$ means $5 \times u$, while $5 + u$ is simply addition. These operations are not the same.

42. Which value of a makes the equation $2a + 3 = 9$ true?

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

Explanation: Substitute $a = 3$: $2(3) + 3 = 6 + 3 = 9$. This satisfies the equation.

43. Which step contains an error in simplifying $3(x + 4) - 2x$?

A) $3x + 12 - 2x$

B) $x + 12$

C) $x + 8$

D) Both A and B are Correct

Answer: C) $x + 8$

Explanation: $3x + 12 - 2x = x + 12$. So $x + 8$ is incorrect.

44. Which of the following simplifications is correct for the expression $(4x+3y)-(3x+4y)$?

A) $x-y$
B) $x+y$
C) $7x+7y$
D) $x-7y$

Answer: A) $x-y$

Explanation: Distribute minus: $4x+3y-3x-4y=x-y$.

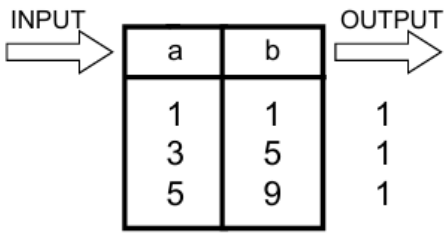
45. Is the simplification of $3j+6k+9h+12$ to $3(j+2k+3h+4)$ valid?

A) Yes, this is correct factoring.
B) No, 3 is not a common factor.
C) Yes, but only if all values are positive.
D) No, it simplifies to something else.

Answer: A) Yes, this is correct factoring.

Explanation: $3(j+2k+3h+4)=3j+6k+9h+12$ by distributive law.

46. Observe the number machine. What is the algebraic expression used in the machine?



A) $a+b$
B) $2a-b$
C) $a-2b$
D) $2b-a$

Answer: B) $2a - b$

Explanation: The output in each case is calculated as twice the first input minus the second.

47. If a tower is built with 'n' layers of blocks, and each layer has 'n' blocks, how many blocks are needed for a tower with 8 layers?

A) $n + 8$
B) $8n$
C) 16
D) 64

Answer: D) 64

Explanation: For a tower with 8 layers, where each layer has 8 blocks, the total number of blocks is $n \times n$ or n^2, which gives $8 \times 8 = 64$.

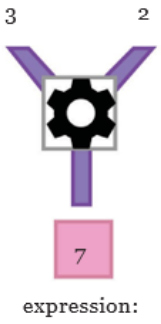
48. A number machine takes two inputs, a and b, and performs this rule: multiply them and then subtract 2. What is the correct expression for the output?

A) $ab + 2$
B) $ab - 2$
C) $a + b - 2$
D) $a - b + 2$

Answer: B) $ab - 2$

Explanation: The rule is: multiply the inputs, then subtract 2 $\rightarrow ab - 2$

49. Which expression matches the outputs in the machine shown?



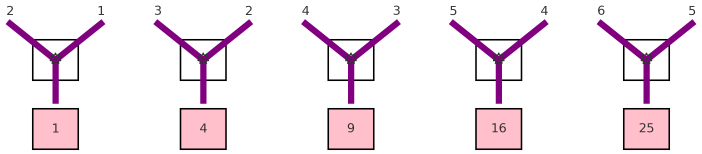
expression:

A) $3a + 2b$
B) $a + b + 5$
C) $3a - b$
D) $a - 3b$

Answer: C) $3a - b$

Explanation: Multiply the first input by 3 and subtract the second: $3a - b$.

50. Determine the expression used in this input-output machine.



A) $2a + b$
B) $a^2 - b$
C) $ab - b$
D) $a(b - 1)$

Answer: C) $ab - b$

Explanation: The operation is: multiply a and b, then subtract b: $ab - b$.

51. Which of the following is a general expression for the sum of two consecutive numbers?

A) $x + x$
B) $x + 2$
C) $x + (x + 1)$
D) $2x + 2$

Answer: C) $x + (x + 1)$

Explanation: Consecutive numbers differ by 1, so if the first is x, the next is $x + 1$.

52. Which algebraic expression represents "triple a number increased by 4"?

A) $3x + 4$
B) $x + 4$
C) $4x + 3$
D) $x + 3$

Answer: A) $3x + 4$

Explanation: "Triple" means multiply by 3, and then increase by 4: $3x + 4$.

53. Which of the following correctly describes the perimeter of a regular polygon with n equal sides each of length s?

A) ns
B) $n + s$
C) $n^2 + s$
D) s^n

Answer: A) ns

Explanation: Perimeter = number of sides × length of one side = ns.

54. What is the difference between $3x$ and $x + x + x$?

A) $x + x + x$ is greater
B) $3x$ is greater
C) They are the same
D) One is addition, the other is subtraction

Answer: C) They are the same

Explanation: $3x$ is shorthand for $x + x + x$; both represent the same quantity.

55. Which statement shows the correct transition from arithmetic to algebraic thinking?

A) $5 + 6 = 11 \Rightarrow x + y = 11$
B) $2 + 2 = 4 \Rightarrow x + x = 2x$
C) $10 \div 2 = 5 \Rightarrow x \div 2 = 5$
D) $7 - 3 = 4 \Rightarrow x - 3 = 4$

Answer: B) $2 + 2 = 4 \Rightarrow x + x = 2x$

Explanation: Replacing equal addends with variables demonstrates transition to algebra: $x + x = 2x$.

56. In the calendar grid below, what is the sum of the diagonally opposite numbers in the 2×2 box starting at number a?

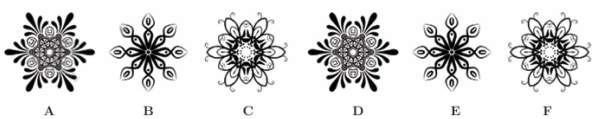


A) $2a$
B) $2a + 8$
C) $2a + 7$
D) $2a + 9$

Answer: B) $2a + 8$

Explanation: The diagonally opposite numbers in the 2×2 box are a and 9, and 2 and 8. Their sums are $a+9$ and $2+8 = 10$, which are equal. So $a+9 = 10$, giving $a = 1$. Substituting into $2a + 8$ gives $2(1) + 8 = 10$, which matches the diagonal sum.

57. The pattern shown represents the border design of a sari. What is the position of the nth appearance of Design C?



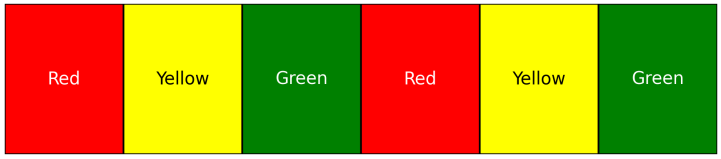
A) $3n$

B) $3n + 1$
C) $3n - 1$
D) $2n + 1$

Answer: A) $3n$

Explanation: Design C occurs at every third position. So, $3n$ gives the position of the nth appearance.

58. Refer to the traffic light sequence in the diagram. What is the color at the 90th position?



A) Red
B) Yellow
C) Green
D) Blue

Answer: C) Green

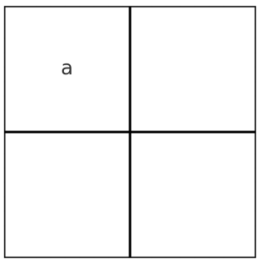
Explanation: If the cycle is Red–Yellow–Green repeating every 3 steps, then $90 \mod 3 = 0$, so it's Green.

59. In the endless calendar row shown, what is the expression for the bottom-right number in terms of top-left number a?



November 2024

Mon	Tue	Wed	Thu	Fri	Sat	Sun
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	32	33	34	35	36	37
38	39	40	41	42	43	44



A) $a + 8$
B) $a + 7$
C) $a + 6$

D) $a + 9$

Answer: A) $a + 8$

Explanation: In a 2×2 grid, bottom-right = top-left $+8$ as per vertical and horizontal shift.

60. In the border pattern below, which design appears at position 122?



A) Design A
B) Design B
C) Design C
D) Cannot be determined

Answer: B) Design B

Explanation: The pattern repeats every 3 designs (A, B, C). To find the design at position 122, we divide 122 by 3. The remainder, 2, corresponds to the second design in the sequence, which is Design B.

61. In the 4-column grid shown below, each new row increases by 4. What is the number in Row 3, Column 2 if Row 1 starts from 1?

1	2	3	4
5	6	7	8
9	?	11	12

A) 6
B) 9
C) 10
D) 8

Answer: C) 10

Explanation: Row 1 has 1–4, Row 2 has 5–8, and Row 3 has 9–12. The second number in Row 3 is 10.

62. In a 4-column grid that begins with 1, what is the number in Row 5, Column 3?

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	?	20

A) 19
B) 20
C) 22
D) 23

Answer: A) 19

Explanation: Each row contains 4 consecutive numbers. The first entry in Row n is $1 + 4(n - 1)$. For Row 5 this is $1 + 4(5 - 1) = 17$. Adding 2 more columns (to reach Column 3) gives $17 + 2 = 19$.

63. Which of these expressions correctly describes the number in Row r, Column c of a 4-column grid that starts from 1?

A) $4r + c$
B) $4(r - 1) + c$
C) $5r + c$
D) $4(r + 1) + c$

Answer: B) $4(r - 1) + c$

Explanation: First number in Row r = $4(r - 1) + 1$; column c = add $c - 1$.

64. If 1 is in the top-left corner of a 4-column grid, what number is in Column 1 of Row 6?

A) 20
B) 21
C) 22
D) 23

Answer: B) 21

Explanation: Row 6 starts from $4 \times (6 - 1) + 1 = 21$

65. In a 4-column grid starting from 1, what row contains number 201? (Use pattern counting)

A) Row 50
B) Row 51
C) Row 49
D) Row 52

Answer: B) Row 51

Explanation: Each row has 4 numbers. Count up by 4s: 201 is the first number in Row 51.

66. What is the algebraic expression for "5 more than a number x"?

A) $5x$
B) $x + 5$
C) $x - 5$
D) $5 - x$

Answer: B) $x + 5$

Explanation: "5 more than" means add 5 to the number: $x + 5$.

67. Which expression represents "4 less than a number a"?

A) $a - 4$
B) $4 - a$
C) $a + 4$
D) $4a$

Answer: A) $a - 4$

Explanation: "4 less than" means subtract 4 from the number: $a - 4$.

68. Write the expression for "2 less than 13 times a number n".

A) $13n + 2$
B) $13n - 2$
C) $2 - 13n$
D) $13 - 2n$

Answer: B) $13n - 2$

Explanation: First multiply by 13, then subtract 2: $13n - 2$.

69. Which of the following means "13 less than 2 times a number y"?

A) $13 - 2y$
B) $2y - 13$
C) $13y - 2$
D) $2y + 13$

Answer: B) $2y - 13$

Explanation: "13 less than" implies subtract 13 from $2y$: $2y - 13$.

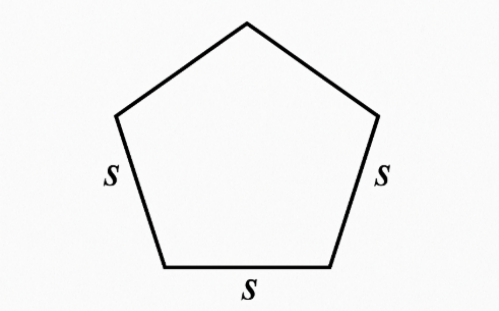
70. What is the correct expression for "a number z divided by 5"?

A) $5z$
B) $z + 5$
C) $\frac{z}{5}$
D) $\frac{5}{z}$

Answer: C) $\frac{z}{5}$

Explanation: The division of a number z by 5 is written as $\frac{z}{5}$.

71. What is the expression for the perimeter of a regular pentagon of side s?

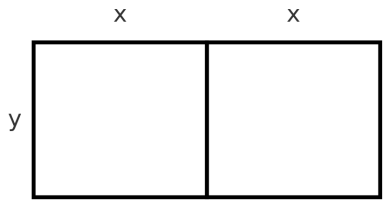


A) $4s$
B) $5s$
C) $s + 2$
D) $2s + 3$

Answer: B) $5s$

Explanation: A regular pentagon has 5 equal sides, so perimeter = $5 \times s = 5s$.

72. The shape below shows two joined rectangles. What is the total area in terms of x and y?

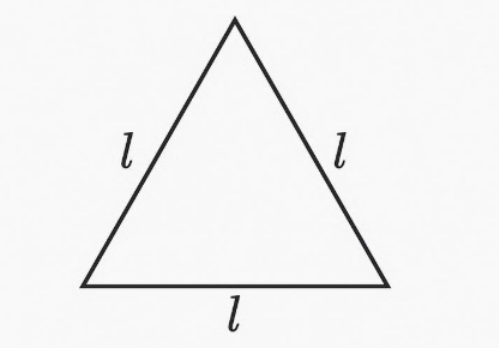


A) $2xy$
B) $x^2 + y^2$
C) $xy + xy$
D) $xy + yx$

Answer: A) $2xy$

Explanation: Each rectangle has area xy. Two such rectangles: $xy + xy = 2xy$.

73. Refer to the triangle below. If each side is of length l, what is the perimeter?



A) $2l$
B) $3l$
C) l^2
D) $l + 2$

Answer: B) $3l$

Explanation: An equilateral triangle has three equal sides: $l + l + l = 3l$.

74. A square has side length a. Another square has side length $a + 2$. What is the difference in their areas?

A) $2a + 2$
B) $4a + 4$
C) $a^2 + 2$
D) $2a + 4$

Answer: B) $4a + 4$

Explanation: Area of the larger square = $(a+2)^2 = a^2 + 4a + 4$; area of smaller square = a^2. So difference = $(a^2 + 4a + 4) - a^2 = 4a + 4$.

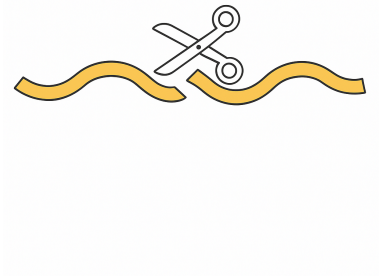
75. A rectangular garden has length $a+3$ and width $b-1$. What is the expression for its area?

A) $ab + 2a - 3b + 3$
B) $ab + 3b - a$
C) $(a + 3)(b - 1)$
D) $ab - 3$

Answer: C) $(a + 3)(b - 1)$

Explanation: Area of a rectangle is length × width. Substituting: $(a + 3)(b - 1)$ gives the required expression.

76. A rope is cut once as shown. How many pieces result from a single cut without folds?

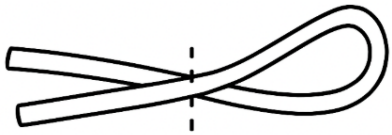


A) 1
B) 2
C) 3
D) 4

Answer: B) 2

Explanation: A single straight cut through an unfolded rope divides it into 2 pieces.

77. If a rope is folded once and cut once, how many pieces are formed?



A) 2
B) 3
C) 4
D) 1

Answer: B) 3

Explanation: A single fold creates overlap. A cut through both layers yields 3 pieces.

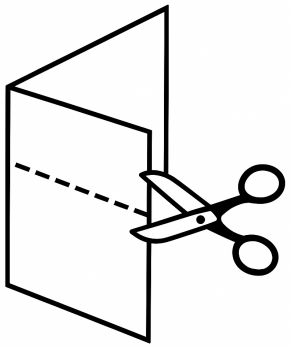
78. A ribbon of length $3x$ is cut into pieces each of length x. How many pieces are formed?

A) x
B) 3
C) $3x$
D) $x + 3$

Answer: B) 3

Explanation: The total length is $3x$, and each piece has length x, so the number of pieces = $3x \div x = 3$.

79. In the folded shape shown, a cut is made as indicated. How many symmetrical shapes will appear when unfolded?



A) 1
B) 2
C) 3
D) 4

Answer: D) 4

Explanation: With a double-fold and symmetric cut, unfolding generates 4 symmetric parts.

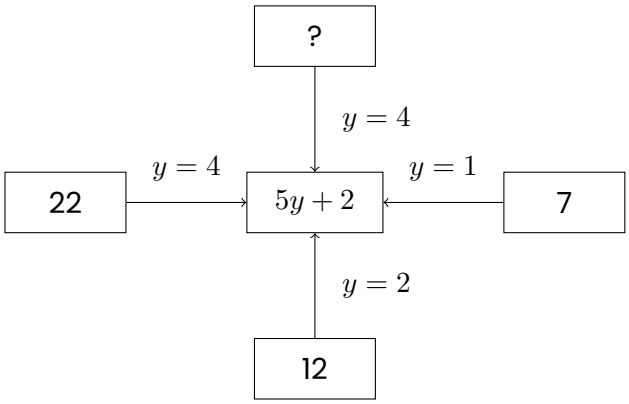
80. Which expression gives the number of pieces if a rope is folded 10 times and cut once?

A) 10
B) 11
C) 20
D) 21

Answer: D) 21

Explanation: The general rule is $2n+1$ pieces for a single cut through n folds. So for $n = 10$, we get $2(10) + 1 = 21$ pieces.

81. In the machine below, each value passes through the rule $5y + 2$. What is the missing output value?

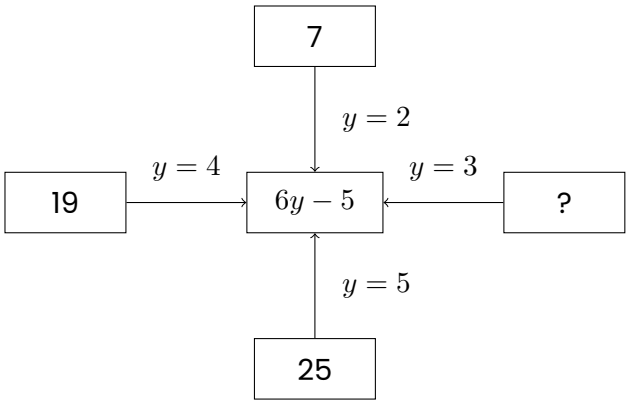


A) 20
B) 22
C) 18
D) 24

Answer: B) 22

Explanation: For $y = 4$:
$5 \cdot 4 + 2 = 20 + 2 = \boxed{22}$

82. Each input passes through the rule $6y - 5$. What is the missing value?



A) 13
B) 18
C) 20
D) 12

Answer: A) 13

Explanation: For $y = 3$:
$6 \cdot 3 - 5 = 18 - 5 = \boxed{13}$

83. What is the nth term expression for the sequence: $3,\ 6,\ 9,\ 12,\ \ldots$**?**

A) $3n$
B) $2n + 1$
C) $n + 2$
D) n^2

Answer: A) $3n$

Explanation: It's a multiple of 3 pattern. nth term = $3 \times n$.

84. Find the 12th term in the sequence: $2.5,\ 2.7,\ 2.9,\ 3.1,\ \ldots$

A) 4.7
B) 4.5
C) 4.9
D) 5.1

Answer: A) 4.7

Explanation: Common difference is 0.2. $T_n = 2.5 + (n-1) \times 0.2 \Rightarrow T_{12} = 4.7$.

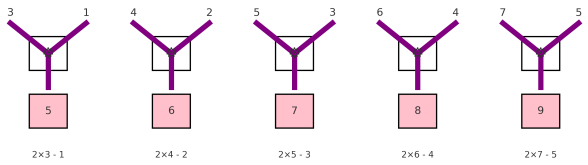
85. What is the general term of the pattern: $1,\ 3,\ 5,\ 7,\ \ldots$**?**

A) $2n + 1$
B) $2n$
C) $2n - 1$
D) n^2

Answer: C) $2n - 1$

Explanation: This is a sequence of odd numbers. General term = $2n - 1$.

86. Observe the number machine below. Which expression gives the output in terms of inputs a and b?



A) $a + b$

B) ab

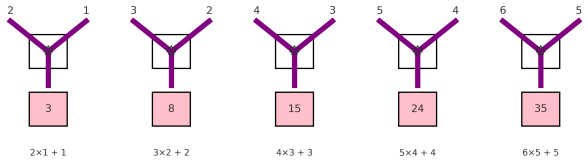
C) $2a - b$

D) $a - 2b$

Answer: C) $2a - b$

Explanation: The output is calculated as twice the first number minus the second: $2a - b$.

87. Find the formula for this number machine and complete the output for inputs a and b:



A) $a + b$

B) $a + 2b$

C) $ab + b$

D) $ab - b$

Answer: C) $ab + b$

Explanation: Multiply inputs, then add the second: $ab + b$.

88. Imagine a repeating pattern of shapes arranged in the order: Circle, Square, Triangle, Circle, Square, Triangle... What is the position of the nth occurrence of the Triangle?

A) 3n

B) 2n+1

C) 3n+1

D) n+3

Answer: A) $3n$

Explanation: The Triangle is the third shape in the repeating sequence of three, so its position is always a multiple of 3, which is represented by the expression $3n$.

89. If Design B appears at positions 2, 5, 8, 11,... what is the formula for the nth appearance of B?

A) $3n - 1$
B) $3n + 1$
C) $3n$
D) $2n + 2$

Answer: A) $3n - 1$

Explanation: Each new position is 3 ahead of the last; start at 2 →formula: $3n - 1$.

90. What is the expression for the total cost of a rope if its length is x metres and it costs ₹12 per metre?

A) $12 + x$
B) $x \div 12$
C) $12x$
D) $x - 12$

Answer: C) $12x$

Explanation: Cost per metre is ₹12, total cost = ₹$12 \times x = 12x$.

91. If $x = 3$, which option correctly evaluates the expression $2x + 5$?

A) 11
B) 9
C) 8
D) 7

Answer: A) 11

Explanation: Substitute $x = 3$: $2 \times 3 + 5 = 6 + 5 = 11$

92. What is the value of $4a - 2$ when $a = 5$?

A) 18
B) 20
C) 22
D) 15

Answer: A) 18

Explanation: $4 \times 5 - 2 = 20 - 2 = 18$

93. Choose the correct value of $5p - 2$ for $p = 6$.

A) 28
B) 30
C) 32
D) 35

Answer: A) 28

Explanation: $5 \times 6 - 2 = 30 - 2 = 28$

94. Find the value of $3x - 2y + 4$ when $x = 2$ and $y = 3$.

A) 4
B) 6
C) 8
D) 10

Answer: A) 4

Explanation: $3 \times 2 - 2 \times 3 + 4 = 6 - 6 + 4 = 4$

95. What is the value of $3x + 4y$ when $x = 2,\ y = 5$?

A) 26
B) 23
C) 22
D) 24

Answer: A) 26

Explanation: $3 \times 2 + 4 \times 5 = 6 + 20 = 26$

96. Simplify the expression: $5x + 3x - 2x$.

A) $6x$
B) $7x$
C) $8x$

D) $10x$

Answer: A) $6x$

Explanation: Combine like terms: $5x + 3x = 8x$, then $8x - 2x = 6x$.

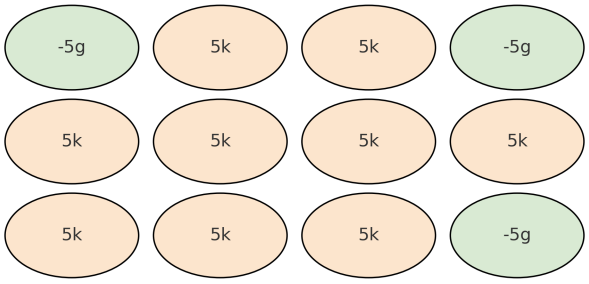
97. What is the simplified form of: $4a - (2a - 3)$**?**

A) $2a - 3$
B) $2a + 3$
C) $6a - 3$
D) $2a$

Answer: B) $2a + 3$

Explanation: Distribute the minus: $4a - 2a + 3 = 2a + 3$

98. What is the simplified form of the last tile group made of $5k, -5g$**?**



A) $25k - 20g$
B) $30k - 25g$
C) $45k - 15g$
D) $40k - 30g$

Answer: C) $45k - 15g$

Explanation:
Seven $5k$ tiles: $9 \times 5k = 45k$
Six $-5g$ tiles: $3 \times (-5g) = -15g$

$$\Rightarrow 45k - 15g$$

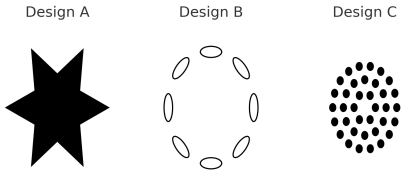
99. Find the simplified form of: $2(x-1)+3(x+2)$.

A) $5x+4$
B) $5x+1$
C) $5x-4$
D) $6x+1$

Answer: A) $5x+4$

Explanation: Expand: $2x-2+3x+6=5x+4$

100. In the border design below, which design appears at position 147?



A) Design A
B) Design B
C) Design C
D) Cannot be determined

Answer: C) Design C

Explanation: A–B–C repeats every 3 positions. 147 is divisible by 3, so it corresponds to Design C.

101. Simplify the expression: $2(n+4)+3(n-1)$.

A) $5n+3$
B) $6n+1$
C) $5n+5$
D) $6n+3$

Answer: C) $5n+5$

Explanation:
First expand the brackets:

$$2(n+4)=2n+8, \quad 3(n-1)=3n-3$$

Now combine:

$$2n+8+3n-3=\boxed{5n+5}$$

102. A fence is built using posts and rails. The first section of fence requires 2 posts and 3 rails. Each additional section requires 1 new post and 3 new rails. What is the expression for the total number of posts and rails combined for 'n' sections of fence?

A) 4n + 1
B) 3n + 2
C) 5n - 1
D) 4n - 1

Answer: A) 4n + 1

Explanation: The first section has 2 posts and 3 rails (total 5). Each additional section adds 1 post and 3 rails (total 4). This forms the expression $4n + 1$. For $n = 1$, it is $4(1) + 1 = 5$. For $n = 2$, it is $4(2) + 1 = 9$ (which is $5 + 4$).

103. The figure below shows a zigzag matchstick pattern. If each "V" uses 2 matchsticks and shares 1 with the next, what is the formula for n Vs?



A) $2n$
B) $n + 1$
C) $n + 2$
D) $n + 3$

Answer: B) $n + 1$

Explanation: First "V" uses 2 matchsticks. Each next "V" shares 1, so adds 1: $2 + (n - 1) = n + 1$.

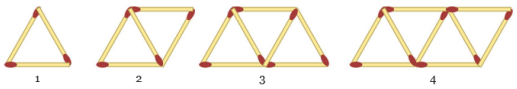
104. A pattern of rectangles is formed using matchsticks. Each rectangle shares one side with the next. What is the total matchsticks used for n rectangles?

A) $4n$
B) $3n + 1$
C) $3n + 2$
D) $4n - 1$

Answer: B) $3n + 1$

Explanation: First rectangle = 4 matchsticks. Each new one adds 3: $4 + 3(n - 1) = 3n + 1$.

105. Refer to the triangular matchstick pattern. If the first triangle uses 3 sticks and each new one shares a side, what is the expression for n triangles?



A) $3n$
B) $2n + 2$
C) $3 + 2(n - 1)$
D) $2n + 3$

Answer: C) $3 + 2(n - 1)$

Explanation: First triangle = 3 sticks; every next adds 2 (shares 1): total = $3 + 2(n - 1)$.

106. In the expression path diagram, the expression $w + 2$ becomes $4w + 20$ on one path. What expression results from the path: $w + 2 \to w + 5 \to \times 4$?

A) $4w + 20$
B) $4w + 8$
C) $4w + 12$
D) $4w + 10$

Answer: A) $4w + 20$

Explanation:
Starting from $w + 2$, the path adds 3:

$$w + 2 + 3 = w + 5$$

Then multiply by 4:

$$4(w + 5) = 4w + 20$$

107. Simplify the expression: $2(r + s + 12)$.

A) $2r + s + 12$

B) $r + s + 24$
C) $2r + 2s + 24$
D) $2r + 2s + 12$

Answer: C) $2r + 2s + 24$

Explanation:
Distribute the 2:

$$2(r + s + 12) = 2r + 2s + 24$$

108. Which of the following expressions is equivalent to $3y + 4y - 2 + 5$?

A) $7y + 3$
B) $7y - 3$
C) $y + 7$
D) $12y - 2$

Answer: A) $7y + 3$

Explanation:
Combine like terms:

$$3y + 4y = 7y, \quad -2 + 5 = 3, \quad \text{so final expression is } 7y + 3$$

109. What is the value of $10y - 3$ when $y = 7$?

A) 67
B) 73
C) 70
D) 77

Answer: A) 67

Explanation: $10 \times 7 - 3 = 70 - 3 = 67$

110. For which value of y do the expressions $6y + 5$ and $3(2y + 1) + 2$ give the same result?

A) $y = 0$
B) $y = 1$
C) $y = 2$
D) $y = 3$

Answer: C) $y = 2$

Explanation:
Check both expressions at $y = 2$: Left: $6 \cdot 2 + 5 = 12 + 5 = 17$ Right: $3(2 \cdot 2 + 1) + 2 = 3(5) + 2 = 15 + 2 = 17$ So both expressions give the same value when $y = 2$.

111. A pen costs ₹12 and a pencil costs ₹5. What expression gives the total cost for x pens and y pencils?

A) $12 + 5 + x + y$
B) $12x + 5y$
C) $x + y + 17$
D) $12xy + 5$

Answer: B) $12x + 5y$

Explanation: Multiply each item's price by quantity: pens ₹12×x, pencils ₹5×y.

112. A taxi charges ₹50 as base fare and ₹15 per km. What expression gives cost for d kilometers?

A) $15d + 50$
B) $50d + 15$
C) $15 + d + 50$
D) $15 + 50d$

Answer: A) $15d + 50$

Explanation: Cost = ₹50 (fixed) + ₹15×distance d.

113. A coconut costs ₹35 and jaggery ₹60/kg. What is the total cost for c coconuts and j kg jaggery?

A) $35c + 60j$
B) $60c + 35j$
C) $c + j + 95$
D) $35 + 60 + cj$

Answer: A) $35c + 60j$

Explanation: Use unit cost: ₹35×c, ₹60×j

114. The cost of 1 roti is ₹8 and 1 pulao is ₹20. What is the expression for x rotis and y pulaos?

A) $8x + 20y$
B) $20x + 8y$
C) $x + y + 28$
D) $8xy + 20$

Answer: A) $8x + 20y$

Explanation: Total = ₹8×rotis + ₹20×pulaos

115. If 1 table costs ₹75 and 1 chair ₹40, what is the expression for t tables and c chairs?

A) $75t + 40c$
B) $40t + 75c$
C) $t + c + 115$
D) $75tc + 40$

Answer: A) $75t + 40c$

Explanation: Multiply unit cost by number of items.

116. A flour mill takes 10 seconds to start and 8 seconds per kg of grain. What is total time for y kg?

A) $10 + 8y$
B) $10y + 8$
C) $8y - 10$
D) $10 + y + 8$

Answer: A) $10 + 8y$

Explanation: Startup time + $y \times 8$ seconds = $10 + 8y$

117. A machine runs for 5 mins setup + 3 mins per unit. What is the expression for n units processed?

A) $5 + 3n$
B) $5n + 3$
C) $3 + 5n$

D) $3n - 5$

Answer: A) $5 + 3n$

Explanation: Fixed setup time + variable time: $5 + 3n$

118. A rope costs ₹12/m. What is cost for a rope of x metres?

A) $12 + x$
B) $x \div 12$
C) $12x$
D) $x - 12$

Answer: C) $12x$

Explanation: ₹12 per metre: cost $= 12x$

119. A plumber charges ₹200 for a visit plus ₹100/hour. What is cost for h hours of work?

A) $200h + 100$
B) $100h + 200$
C) $h + 200$
D) $300h$

Answer: B) $100h + 200$

Explanation: ₹100/hour + ₹200 base fee = $100h + 200$

120. A train stops 3 times and travels 4 segments. If stop = 2 mins and segment = t mins, what is total time?

A) $3 + 4t$
B) $4 + 3t$
C) $3t + 4$
D) $4t + 6$

Answer: D) $4t + 6$

Explanation: 3 stops = 6 mins; 4 segments = $4t$ mins $\rightarrow$total = $4t + 6$

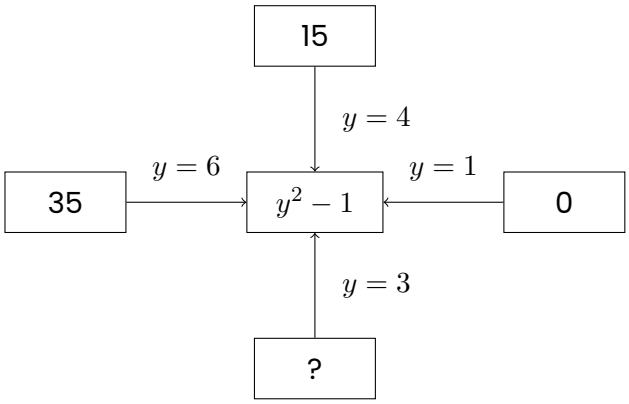
121. If a rectangle has a length of $x + 5$ and a width of x, what is its perimeter?

A) 2x + 5
B) 2x + 10
C) 4x + 10
D) $x^2 + 5x$

Answer: C) 4x + 10

Explanation: The perimeter of a rectangle is $P = 2(L + W)$. Substituting the given values, we get $P = 2((x + 5) + x)$, which simplifies to $P = 2(2x + 5)$ or $4x + 10$.

122. The machine uses the rule $y^2 - 1$. What is the missing output value?

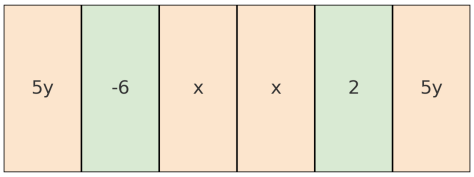


A) 10
B) 9
C) 8
D) 7

Answer: C) 8

Explanation:
For $y = 3$:
$3^2 - 1 = 9 - 1 = \boxed{8}$

123. Which of the following is the simplified expression formed by the blocks in the first group?



A) $10y + 2x - 4$
B) $10y + 2x - 6$
C) $10y + 2x + 6$
D) $5y + x - 4$

Answer: B) $10y + 2x - 6$

Explanation:
There are two $5y$ tiles: $5y + 5y = 10y$
Two x tiles: $x + x = 2x$
Two numbers: $-6 + 2 = -4$

$$\Rightarrow 10y + 2x - 4$$

124. Simplify the expression: $ab + bc + ab$**.**

A) $2ab + bc$
B) $ab + bc$
C) $2ab + 2bc$
D) $ab + ab + bc$

Answer: A) $2ab + bc$

Explanation:
Combine like terms:

$$ab + ab = 2ab, \quad \text{so total becomes } 2ab + bc$$

125. Simplify the expression: $4w - (w - 1)$**.**

A) $4w - w + 1 = 3w + 1$
B) $4w - w - 1 = 3w - 1$
C) $3w + 1$
D) $3w - 1$

Answer: C) $3w + 1$

Explanation:
First expand:

$$4w - (w - 1) = 4w - w + 1 = 3w + 1$$

126. The large rectangle is split into two smaller rectangles. What will be the expression for the total area?

A) $ab + ac$
B) $a + b + c$
C) $a(b + c)$
D) Both A and C

Answer: D) Both A and C

Explanation: By distributive law: $ab + ac = a(b + c)$

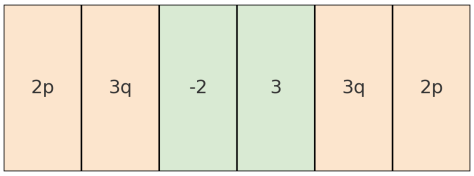
127. A rectangle has length 12 and is split into parts of 4 and 8. What is the expression for area with width n?

A) $4n + 8n$
B) $12n$
C) $n(4 + 8)$
D) All of the above

Answer: D) All of the above

Explanation: $4n + 8n = 12n = n(4 + 8)$

128. From the group with $2p, 3q, 3, -2$, what is the simplified form of the expression?



A) $4p + 6q + 1$
B) $6p + 6q + 1$
C) $4p + 6q + 5$
D) $6p + 4q + 1$

Answer: A) $4p + 6q + 1$

Explanation:
Two $2p$ tiles: $2p + 2p = 4p$

Two $3q$ tiles: $3q + 3q = 6q$
Two numbers: $3 + (-2) = 1$

$$\Rightarrow 4p + 6q + 1$$

129. Simplify the expression: $6v + 2v - v$**.**

A) $9v$
B) $7v$
C) $8v$
D) $6v$

Answer: B) $7v$

Explanation:
Combine the like terms:

$$6v + 2v - v = 8v - v = 7v$$

130. A rectangular carpet has a total length of 12 metres. A border of 4 metres is trimmed off. If the width is n metres, what is the expression for the remaining area of the carpet?

A) $8n$
B) $12n$
C) $4n$
D) $n + 8$

Answer: A) $8n$

Explanation:
Remaining length after removing border = $12 - 4 = 8$
Area of rectangle = length × width = $8n$

131. Simplify: $7p - 2p + 4q + q$

A) $9p + 5q$
B) $5p + 5q$
C) $9p + q$
D) $5p + 4q$

Answer: B) $5p + 5q$

Explanation: Combine like terms: $7p - 2p = 5p$, $4q + q = 5q$

132. What is the simplified form of $3x + 5 - x + 7$?

A) $2x + 12$
B) $4x + 2$
C) $2x + 2$
D) $3x + x + 12$

Answer: A) $2x + 12$

Explanation: Combine: $3x - x = 2x$; $5 + 7 = 12$

133. Simplify the expression: $4a + 3b - 2a + 6b$

A) $2a + 9b$
B) $6a + 3b$
C) $2a + 3b$
D) $4a + 9b$

Answer: A) $2a + 9b$

Explanation: Combine like terms: $4a - 2a = 2a$, $3b + 6b = 9b$

134. Remove brackets and simplify: $2(x + 4) + 3x$

A) $2x + 4 + 3x$
B) $5x + 4$
C) $2x + 8 + 3x$
D) $5x + 8$

Answer: D) $5x + 8$

Explanation: Distribute: $2x + 8 + 3x = 5x + 8$

135. Simplify: $6p - (2p + 5)$

A) $4p - 5$
B) $8p + 5$
C) $4p + 5$

D) $6p + 2p + 5$

Answer: A) $4p - 5$

Explanation: Open bracket with minus: $6p - 2p - 5 = 4p - 5$

136. Which expression represents "Add 4 to a number and multiply the result by 3"?

A) $4 + 3x$
B) $3(x + 4)$
C) $3x + 4$
D) $(x + 3) \times 4$

Answer: B) $3(x + 4)$

Explanation: Do addition first: $x + 4$, then multiply entire result by 3: $3(x + 4)$

137. What is the simplified form of 3(2x+5) - 4(x-2)?

A) 2x+7
B) 2x+13
C) 10x+7
D) 2x+23

Answer: D) 2x+23

Explanation: First, we apply the distributive property to both terms: $3(2x+5) = 6x + 15$ and $-4(x - 2) = -4x + 8$. Then, we combine the like terms: $(6x - 4x) + (15 + 8) = 2x + 23$.

138. Choose the correct expression for "Subtract 7 from a number and multiply by 2".

A) $2x - 7$
B) $2(x - 7)$
C) $x - 7 + 2$
D) $(7 - x) \times 2$

Answer: B) $2(x - 7)$

Explanation: Subtract first: $x - 7$, then multiply: $2(x - 7)$

139. What is the result of applying distributive property to $3(x+2)+4x$?

A) $3x+6+4x$
B) $7x+2$
C) $3x+2+4x$
D) $3x+4x+2$

Answer: A) $3x+6+4x$

Explanation: Distribute: $3\times x=3x$, $3\times 2=6$; then add $4x$

140. Which of the following correctly expands $2(3x+5)$?

A) $6x+5$
B) $6x+10$
C) $3x+7$
D) $2x+15$

Answer: B) $6x+10$

Explanation: Multiply each term by 2: $2\times 3x=6x$, $2\times 5=10$

141. A student simplified $4x+3x-5$ as $7x-5$. Is the simplification correct?

A) Yes, correct
B) No, it should be $7x+5$
C) No, it should be $4x+3x-5x$
D) No, terms are not like terms

Answer: A) Yes, correct

Explanation: $4x+3x=7x$, so $7x-5$ is correctly simplified.

142. In the expression below, the student made a mistake in removing brackets. What is the correct result?

Given: $2(x+4)+3x$

A) $5x+8$
B) $3x+4$
C) $5x+4$
D) $3x+8$

Answer: A) $5x + 8$

Explanation:
Distribute first: $2(x + 4) = 2x + 8$, then add $3x$:

$$2x + 8 + 3x = 5x + 8$$

143. A student simplified $3(a + 5)$ as $3a + 5$. What was the error?

A) Missed multiplying 5
B) Used wrong operation
C) Reversed terms
D) No error

Answer: A) Missed multiplying 5

Explanation: $3(a + 5) = 3a + 15$, not $3a + 5$

144. Two expressions are compared below. What is the error in the second simplification?

Original: $6x - (3x - 5)$ Wrong: $6x - 3x - 5$

A) Subtracted instead of added
B) Incorrect bracket removal
C) Ignored sign of the term
D) All of the above

Answer: D) All of the above

Explanation: Common simplification errors: incorrect signs, bracket misuse, and adding/subtracting incorrectly.

145. A student simplified $6p - (2p + 4)$ as $6p - 2p + 4$. What is the mistake?

A) Did not apply negative to both terms
B) Reversed subtraction
C) Incorrect addition

D) Treated it as multiplication

Answer: A) Did not apply negative to both terms

Explanation: Correct: $6p - 2p - 4$; student added 4 instead of subtracting.

146. Simplify the expression: $3(2x + 1) + x$**.**

A) $6x + 1$
B) $7x + 3$
C) $5x + 3$
D) $6x + x + 1$

Answer: B) $7x + 3$

Explanation:

$$3(2x + 1) = 6x + 3, \quad \text{then add } x: \quad 6x + 3 + x = 7x + 3$$

147. A pizza costs ₹240. Ravi pays with r ₹100-notes. What expression shows his balance amount?

A) $100r - 240$
B) $240 - 100r$
C) $240r - 100$
D) $r - 240$

Answer: A) $100r - 240$

Explanation: Total paid = ₹$100r$, subtract cost: balance = $100r - 240$

148. If a matchstick pattern uses $4n - 1$ sticks for n squares in a line, how many are used for 10 squares?

A) 39
B) 40
C) 41
D) 35

Answer: A) 39

Explanation: Use formula: $4n - 1 = 4 \times 10 - 1 = 39$

149. In a 3×3 grid, the value of each cell at row i and column j is given by the expression $4i - j$. What is the value of the cell in row 3, column 2?

A) 10
B) 12
C) 14
D) 15

Answer: A) 10

Explanation: Substituting $i = 3$ and $j = 2$ into the expression $4i - j$ gives $4(3) - 2$, which simplifies to $12 - 2 = 10$.

150. A pack of pens costs ₹45. What is the total cost for x packs and 3 extra pens (₹15 each)?

A) $45x + 45$
B) $45x + 3$
C) $45x + 15x$
D) $45x + 45x$

Answer: A) $45x + 45$

Explanation: ₹45 per pack, ₹15 per pen × 3 pens = ₹45. Total = $45x + 45$

Two-Mark Problems

151. Write an algebraic expression for "4 more than thrice a number x".

Solution:

Given: 4 more than thrice a number x.

To Find : An algebraic expression

Let x be a number.

i.e., Thrice a number x means $3x$.

So, 4 more than means we add 4 to $3x$.

Which gives $\Rightarrow 3x + 4$.

Hence, the required expression is:

$$3x + 4$$

Answer: $3x + 4$

Keywords: $3x$, $3x + 4$

152. A pen costs ₹12 and a pencil costs ₹5. Write an expression for the total cost of p pens and q pencils.

Solution:

Given: Cost of one pen = ₹12 .
cost of one pencil = ₹5

To Find: An expression for total cost of p pens and q pencils

Let the number of pens be p and pencils be q.

Cost of p pens = $12 \times p = 12p$

Cost of q pencils = $5 \times q = 5q$

Total cost = $12p + 5q$

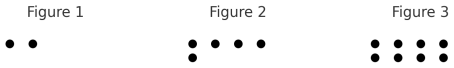
Hence, the required expression is:

$$12p + 5q$$

Answer: $\boxed{12p + 5q}$

Keywords: $12p$, $5q$, $12p + 5q$

153. Look at the pattern of dots below. Write an expression for the number of dots in the nth figure.



Solution:

Given: A growing dot pattern.

To Find: An expression for the number of dots in the nth figure.

Observe:
Figure 1 has 2 dots.
Figure 2 has 5 dots.
Figure 3 has 8 dots.

We see that the number of dots increases by 3 each time.

We can build the pattern as:
Figure 1 = $3 \times 1 - 1 = 2$
Figure 2 = $3 \times 2 - 1 = 5$
Figure 3 = $3 \times 3 - 1 = 8$

So, the number of dots in the nth figure is:

$$3n - 1$$

Answer: $\boxed{3n - 1}$

Keywords: 2, 5, 8, $3n - 1$

154. A rickshaw charges ₹30 for the first kilometre and ₹12 for every additional kilometre. Write an expression for the total fare for x kilometres.

Solution:

Given: ₹30 for the first kilometre and ₹12 for each additional kilometre.

To Find: An algebraic expression for total fare when the distance is x km.

If $x = 1$, the fare is ₹30.

If $x > 1$, then first km = ₹30, and remaining $x - 1$ km are charged ₹12 per km.

So, extra fare = $12 \times (x - 1) = 12(x - 1)$

Total fare = ₹30 + ₹$12(x - 1)$ = $30 + 12(x - 1)$

Hence, the required expression is:

$$30 + 12(x - 1)$$

Answer: $\boxed{30 + 12(x - 1)}$

Keywords: $x - 1$, $12(x - 1)$, $30 + 12(x - 1)$

155. A restaurant charges ₹80 for a table and ₹25 per person for food. Write an expression for the total cost if p people book a table. Also, find the cost if 6 people come.

Solution:

Given: Table charge = ₹80, Food per person = ₹25, Number of people = p

To Find: Expression for total cost and evaluate for $p = 6$

Step 1: Write the expression:

$$\text{Total cost} = 80 + 25p$$

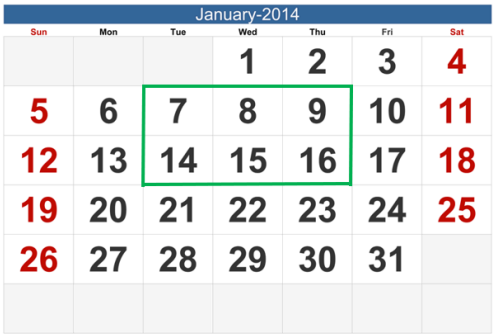
Step 2: Substitute $p = 6$:

$$80 + 25 \times 6 = 80 + 150 = 230$$

Answer: ₹230

Keywords: $80 + 25p,\ 25 \times 6,\ 230$

156. In a calendar month, if any 2×3 grid full of dates is chosen as shown below, write expressions for all the blank cells in terms of w, where the bottom middle cell is w.



Solution:

Given: A calendar grid with 2 rows and 3 columns.

To Find: Algebraic expressions for each cell in terms of bottom-middle cell w.

Let the bottom middle cell be w.
Since the grid is from a calendar, each day increases by 1 from left to right, and by 7 from bottom to top.

So the bottom row values are: Left = $w - 1$, Middle = w, Right = $w + 1$

Top row values are: Left = $w - 8$, Middle = $w - 7$, Right = $w - 6$

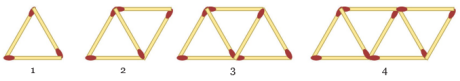
Hence, the completed grid expressions are:

$w - 8$	$w - 7$	$w - 6$
$w - 1$	**w**	$w + 1$

Answer: All cells in terms of w as above

Keywords: $w-8$, $w-7$, $w-6$, $w-1$, w, $w+1$

157. Observe the pattern below formed with matchsticks. Write an expression for the number of matchsticks used to form n triangles in a row.



Solution:

Given: A growing pattern of triangles using matchsticks.

To Find: Expression for the total number of matchsticks in n triangles placed side by side.

Observe:
- 1 triangle uses 3 matchsticks.
- 2 triangles share one side and use 5 matchsticks.
- 3 triangles share sides and use 7 matchsticks.

We see that each additional triangle after the first adds 2 matchsticks. So, total matchsticks = $3+2(n-1)$

Simplify:

$$3+2(n-1)=3+2n-2=2n+1$$

Hence, the required expression is:

$$2n+1$$

Answer: $\boxed{2n+1}$

Keywords: 3, $2(n-1)$, $2n+1$

158. Each letter "F" in the pattern below is made using 4 matchsticks. Write an expression for the total number of matchsticks required to make n letters "F".

F F F

Solution:

Given: Each letter "F" is constructed with 4 matchsticks.

To Find: An algebraic expression for total matchsticks to make n letters.

Observe:
- 1 letter uses 4 matchsticks.
- 2 letters use $2 \times 4 = 8$ matchsticks.
- 3 letters use $3 \times 4 = 12$ matchsticks.

So the pattern follows multiplication:
Total matchsticks = $4 \times n$

Hence, the required expression is:

$$4n$$

Answer: $\boxed{4n}$

Keywords: 4, $4n$

159. If $a = -4$, then is the expression $10 - a = 6$ correct? If not, identify the mistake and write the correct value.

Solution:

Given: $a = -4$ and the expression is $10 - a$.

To Check: Whether $10 - a = 6$ is true.

Substitute $a = -4$ into the expression:

$$10 - (-4) = 10 + 4 = 14$$

The mistake: Negative sign was not handled correctly — subtracting a negative is actually adding.

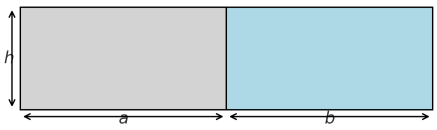
Hence, the corrected value of the expression is:

$$10 - a = 14$$

Answer: $\boxed{14}$

Keywords: $10 - (-4)$, $10 + 4$, 14

160. A rectangle is split into two parts along its length as shown below. If one part has length a and the other has length b, and the height is h, write an expression for the total area of the rectangle.



Solution:

Given: A rectangle is split into two parts. One part has length a, the other part has length b. Both parts have height h.

To Find: Expression for the total area of the rectangle.

Area of left part = $a \times h = ah$
Area of right part = $b \times h = bh$

Total area = $ah + bh$

Factor using distributive law:

$$ah + bh = h(a + b)$$

Hence, the required expression for total area is:

$$h(a + b)$$

Answer: $\boxed{h(a + b)}$

Keywords: ah, bh, $ah + bh$, $h(a + b)$

161. Are the expressions $5u$ and $5 + u$ equal for all values of u? Use the values $u = 2,\ 5,\ 8,\ 11$ to check and compare the outputs.

Solution:

Given: Two expressions: $5u$ and $5 + u$.

To Check: Whether they give equal values for different values of u.

Evaluate for each value of u:

u	$5u$	$5+u$
2	10	7
5	25	10
8	40	13
11	55	16

We can clearly see that $5u \neq 5+u$ for all values. So the expressions are not equal.

Answer: Not equal

Keywords: $5 \times 2 = 10$, $5+2=7$, all pairs different

162. Are the expressions $3(y+2)$ and $3y+2$ equal? Use the values $y = 2,\ 4,\ 6,\ 8$ to compare the outputs.

Solution:

Given: Two expressions: $3(y+2)$ and $3y+2$.

To Find: Whether they are equal for different values of y.

Evaluate for each value of y:

y	$3(y+2)$	$3y+2$
2	$3 \times 4 = 12$	$6+2=8$
4	$3 \times 6 = 18$	$12+2=14$
6	$3 \times 8 = 24$	$18+2=20$
8	$3 \times 10 = 30$	$24+2=26$

The values are not the same in any case. So, the two expressions are not equal.

Answer: Not equal

Keywords: $3(y+2)$, $3y+2$, evaluated values, comparison

163. A person earns ₹200 per day. On Sunday, he gets a bonus of ₹100. Write an expression for his total earnings in d days including one Sunday.

Solution:

Given: Daily wage = ₹200, Bonus on Sunday = ₹100

To Find: Total earnings for d days including one Sunday.

Regular earning for d days = ₹$200 \times d = 200d$

Add Sunday bonus = ₹100

Total earnings = $200d + 100$

Hence, the required expression is:

$$200d + 100$$

Answer: $\boxed{200d + 100}$

Keywords: $200d$, 100, $200d + 100$

164. A taxi service charges ₹50 as base fare and ₹18 for every kilometre travelled. Write an expression for the total fare for x kilometres.

Solution:

Given: Base fare = ₹50, Rate per km = ₹18

To Find: Total fare for x kilometres.

Fare for x km = ₹$18 \times x = 18x$
Add base fare ₹50

Total fare = $18x + 50$

Hence, the required expression is:

$$18x + 50$$

Answer: $\boxed{18x + 50}$

Keywords: $18x$, 50, $18x + 50$

165. A carpenter charges ₹600 for making a table and ₹250 for a chair. Write an expression for the total cost of making t tables and c chairs.

Solution:

Given: Table cost = ₹600, Chair cost = ₹250

To Find: Total cost for t tables and c chairs.

Cost of t tables = $600 \times t = 600t$
Cost of c chairs = $250 \times c = 250c$

Total cost = $600t + 250c$

Hence, the required expression is:

$$600t + 250c$$

Answer: $\boxed{600t + 250c}$

Keywords: $600t$, $250c$, $600t + 250c$

166. Write an expression for "twice the sum of a number x and 9", and simplify it.

Solution:

Given: The phrase is "twice the sum of a number x and 9".

To Find: An expression representing this phrase and simplify it.

Step 1: Sum of x and 9 = $x + 9$
Step 2: Twice of that sum = $2(x + 9)$
Step 3: Simplify using distributive law:

$$2(x + 9) = 2x + 18$$

Hence, the required expression is:

$$2x + 18$$

Answer: $\boxed{2x + 18}$

Keywords: $x + 9$, $2(x + 9)$, $2x + 18$

167. Write an algebraic expression for "a number y decreased by the sum of 6 and another number z".

Solution:

Given: The phrase is "a number y decreased by the sum of 6 and another number z".

To Find: An algebraic expression representing the phrase.

Step 1: The sum of 6 and z = $6 + z$
Step 2: y decreased by that sum = $y - (6 + z)$
Step 3: Remove brackets (optional):

$$y - (6 + z) = y - 6 - z$$

Hence, the required expression is:

$$y - 6 - z$$

Answer: $\boxed{y-6-z}$

Keywords: $6+z$, $y-(6+z)$, $y-6-z$

168. Write an expression for "5 less than the product of a number a and 9", and simplify it.

Solution:

Given: The phrase is "5 less than the product of a number a and 9".

To Find: An algebraic expression for this situation and simplify it.

Step 1: Product of 9 and a = $9a$
Step 2: 5 less than that product = $9a-5$

No further simplification is needed as it's already in simplest form.

Hence, the required expression is:

$$9a-5$$

Answer: $\boxed{9a-5}$

Keywords: $9a$, $9a-5$

169. Use the distributive property to expand and simplify the expression: $7(x+5)$**.**

Solution:

Given: Expression is $7(x+5)$

To Find: Expanded and simplified form using distributive property.

Step 1: Apply distributive law:

$$7(x+5)=7\times x+7\times 5$$

Step 2: Simplify each term:

$$7x+35$$

Hence, the simplified expression is:

$$7x+35$$

Answer: $\boxed{7x+35}$

Keywords: $7x$, 35, $7x+35$

170. Expand and simplify the expression: $4(3x - 2)$**.**

Solution:

Given: Expression is $4(3x - 2)$

To Find: Expanded and simplified form using distributive property.

Step 1: Apply distributive property:

$$4(3x - 2) = 4 \times 3x - 4 \times 2$$

Step 2: Simplify each part:

$$12x - 8$$

Hence, the simplified expression is:

$$12x - 8$$

Answer: $\boxed{12x - 8}$

Keywords: $12x$, 8, $12x - 8$

171. Reema is 4 years older than her brother Aryan. Write an expression for Reema's age in terms of Aryan's age. Find Reema's age when Aryan is 11 years old.

Solution:

Given: Aryan's age is a years.

To Find: Expression for Reema's age and her actual age when $a = 11$.

Step 1: Form the expression:

$$\text{Reema's age} = a + 4$$

Step 2: Substitute $a = 11$:

$$a + 4 = 11 + 4 = 15$$

Answer: $\boxed{15 \text{ years}}$

Keywords: $a + 4,\ 11 + 4,\ 15$

172. A pattern is made using L-shaped matchsticks. Each 'L' uses 3 matchsticks and shares one matchstick with the next. Write an expression for the total number of matchsticks used in terms of number of L-shapes n. Find the number of matchsticks used for 5 L-shapes.

Solution:

Given: Each L-shape adds 2 new matchsticks after the first L.

To Find: Expression for total matchsticks and value for $n = 5$.

Step 1: General expression:

$$\text{Total matchsticks} = 3 + 2(n - 1)$$

Step 2: Simplify expression:

$$= 2n + 1$$

Step 3: Substitute $n = 5$:

$$2 \times 5 + 1 = 10 + 1 = 11$$

Answer: $\boxed{11 \text{ matchsticks}}$

Keywords: $2n + 1,\ 2 \times 5 + 1,\ 11$

173. The cost of 1 coconut is ₹35 and the cost of 1 kg jaggery is ₹60. Write an expression for the total cost of buying c coconuts and j kg of jaggery. Also, find the total cost when $c = 2$ and $j = 3$.

Solution:

Given: 1 coconut = ₹35, 1 kg jaggery = ₹60

To Find: Expression and value for total cost at given values of c and j

Step 1: Write the expression:

$$\text{Total cost} = 35c + 60j$$

Step 2: Substitute $c = 2, j = 3$:

$$35 \times 2 + 60 \times 3 = 70 + 180 = 250$$

Answer: ₹250

Keywords: $35c + 60j,\ 70 + 180,\ ₹250$

174. Write an expression for the perimeter of a regular polygon with s sides, each of length l. Also, find the perimeter of a regular hexagon with side length 9 cm.

Solution:

Given: Polygon with s sides, each of length l

To Find: Expression and value for perimeter when $s = 6,\ l = 9$

Step 1: General expression for perimeter:

$$\text{Perimeter} = s \times l$$

Step 2: Substitute $s = 6, l = 9$:

$$6 \times 9 = 54$$

Answer: 54 cm

Keywords: $s \times l,\ 6 \times 9,\ 54$

175. A flour mill takes 10 minutes to grind wheat and 8 minutes for each additional item. Write an expression for total time taken to grind wheat and y other items. Also, find the time taken if $y = 4$.

Solution:

Given: Wheat takes 10 minutes, each other item takes 8 minutes

To Find: Expression for total time, then calculate when $y = 4$

Step 1: Expression for time:

$$\text{Total time} = 10 + 8y$$

Step 2: Substitute $y = 4$:

$$10 + 8 \times 4 = 10 + 32 = 42$$

Answer: $\boxed{42 \text{ minutes}}$

Keywords: $10 + 8y,\ 8 \times 4,\ 42$

176. The letters A, B, C repeat in a pattern as shown. Write an expression to find the letter at the nth position. Which letter is at the 50th position?

A	B	C	A	B	C

Solution:

Given: Pattern repeats every 3 positions (A, B, C)

To Find: General rule using modulo and find letter at position 50

Step 1: Use remainder when $n \div 3$: - If remainder is 1 $\rightarrow$A - If remainder is 2 $\rightarrow$B - If remainder is 0 $\rightarrow$C

Step 2: $50 \div 3 = 16$ remainder 2 $\rightarrow$Position 50 = B

Answer: $\boxed{B}$

Keywords: $n \mod 3,\ 50 \div 3 = 16\ R2,\ B$

177. Simplify the expression: $5x - [3x - 2(x - 4)]$

Solution:

Given: Expression is $5x - [3x - 2(x - 4)]$

To Find: Fully simplified form.

Step 1: Expand the inner bracket:

$$2(x - 4) = 2x - 8$$

Step 2: Substitute into main expression:

$$5x - [3x - (2x - 8)]$$

Step 3: Remove inner bracket:

$$5x - [3x - 2x + 8] = 5x - [x + 8]$$

Step 4: Final simplification:

$$5x - x - 8 = 4x - 8$$

Answer: $\boxed{4x - 8}$

Keywords: $2x - 8,\ x + 8,\ 5x - x - 8,\ 4x - 8$

178. Evaluate the expression $2a - 3b + 4$ **for** $a = 5,\ b = 2$**.**

Solution:

Given: $a = 5,\ b = 2$

To Find: Value of $2a - 3b + 4$

Step 1: Substitute values into expression:

$$2 \times 5 - 3 \times 2 + 4$$

Step 2: Simplify each term:

$$10 - 6 + 4 = 8$$

Answer: $\boxed{8}$

Keywords: $2 \times 5,\ 3 \times 2,\ 10 - 6 + 4,\ 8$

179. A student simplified $4x - [2x + 3(x - 2)]$ and got $4x - 2x + 3x - 6 = 5x - 6$. Spot the mistake and write the correct answer.

Solution:

Given: Expression is $4x - [2x + 3(x - 2)]$

To Find: Identify mistake and simplify correctly.

Step 1: Expand inside the brackets:

$$3(x - 2) = 3x - 6 \Rightarrow 2x + 3x - 6 = 5x - 6$$

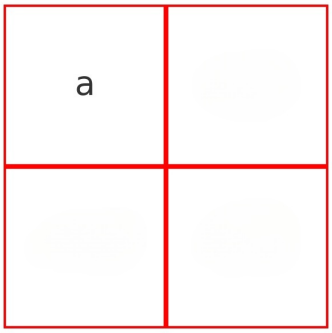
Step 2: Now apply subtraction to the entire bracket:

$$4x - [5x - 6] = 4x - 5x + 6 = -x + 6$$

Answer: $\boxed{-x + 6}$

Keywords: $3x - 6,\ 5x - 6,\ 4x - 5x + 6,\ -x + 6$

180. In a calendar grid, a 2×2 square has the top-left number as a. Fill in the remaining numbers, then find the sum of both diagonals. What do you observe?



Solution:

Given: Top-left value = a, rest follow normal calendar increment pattern

To Find: Expressions and diagonal sums

Step 1: Fill values:

a	$a+1$
$a+7$	$a+8$

Step 2: Diagonal 1 = $a + (a+8) = 2a+8$
Step 3: Diagonal 2 = $(a+1) + (a+7) = 2a+8$

Both diagonals have the same sum.

Answer: $\boxed{2a+8 \text{ (same for both diagonals)}}$

Keywords: $a+8,\ a+1+a+7,\ 2a+8$

181. In a quiz, each correct answer scores 4 points and each wrong answer deducts 1 point. Write an expression for the total score of a student who gives c correct answers and w wrong answers. Also, find the score if $c=7,\ w=3$.

Solution:

Given: +4 for correct answer, –1 for wrong answer

To Find: Expression and total score for $c=7,\ w=3$

Step 1: Expression for score:

$$4c - w$$

Step 2: Substitute values:

$$4 \times 7 - 3 = 28 - 3 = 25$$

Answer: $\boxed{25 \text{ points}}$

Keywords: $4c-w,\ 4\times 7-3,\ 25$

182. Simplify the expression: $7x - 3y + 4x + 6y - 2x$

Solution:

Given: Expression is $7x - 3y + 4x + 6y - 2x$

To Find: Simplified expression by combining like terms.

Step 1: Group like terms:

$$(7x + 4x - 2x) + (-3y + 6y)$$

Step 2: Simplify:

$$9x + 3y$$

Answer: $\boxed{9x + 3y}$

Keywords: $7x + 4x - 2x,\ -3y + 6y,\ 9x + 3y$

183. Write an expression for the perimeter of a rectangle with length l and breadth b. Also, find the perimeter when $l = 10$ cm, $b = 6$ cm.

Solution:

Given: Length = l, Breadth = b

To Find: Expression and value of perimeter when $l = 10,\ b = 6$

Step 1: Use rectangle perimeter formula:

$$\text{Perimeter} = 2(l + b)$$

Step 2: Substitute values:

$$2(10 + 6) = 2 \times 16 = 32$$

Answer: $\boxed{32 \text{ cm}}$

Keywords: $2(l + b),\ 2 \times 16,\ 32$

184. Evaluate the expression $3x - 2y + 5$ when $x = 4,\ y = 1$.

Solution:

Given: $x = 4,\ y = 1$

To Find: Value of $3x - 2y + 5$

Step 1: Substitute values:

$$3 \times 4 - 2 \times 1 + 5$$

Step 2: Simplify each term:

$$12 - 2 + 5 = 15$$

Answer: $\boxed{15}$

Keywords: $3 \times 4,\ 2 \times 1,\ 12 - 2 + 5,\ 15$

185. In a calendar grid, a 3×3 square has the top-left value as x. Write expressions for all other entries and find the sum of all rows, columns, and diagonals. What do you observe?

Solution:

Step 1: Complete the grid with expressions. The numbers in a calendar grid increase by 1 horizontally and by 7 vertically.

x	$x+1$	$x+2$
$x+7$	$x+8$	$x+9$
$x+14$	$x+15$	$x+16$

Step 2: Find the sum of the diagonals.

Main Diagonal (top-left to bottom-right): $x + (x+8) + (x+16) = 3x + 24$
Other Diagonal (top-right to bottom-left): $(x+2)+(x+8)+(x+14) = 3x+24$

Observation The sum of both diagonals is the same, which is $3x + 24$.

Answer: $\boxed{3x + 24 \text{ (same for both diagonals)}}$

Keywords: 3×3 square, $x + 8$, $x + 1, x + 7$, $3x + 24$, Diagonal sum

Five-Mark Problems

186. In a calendar-like number grid, a 2 × 2 square is formed with top-left number as 'a'.

(a) **Write algebraic expressions for all the numbers in the 2×2 square.**

(b) **Find the sum of both diagonals. Simplify and compare.**

(c) **Verify your result by substituting $a = 15$.**

(d) **Explain in words why the diagonal sums will always be equal for any a.**

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24	25	26	27	28	29	30
31	32	33	34	35	36	37
38	39	40	41	42	43	44
⋮	⋮	⋮	⋮	⋮	⋮	⋮

a	?
?	?

Solution:

(a) The four positions in the 2×2 square are:

a	$a+1$
$a+7$	$a+8$

(b) Diagonal 1: $a + (a + 8) = 2a + 8$
Diagonal 2: $(a+1)+(a+7) = 2a+8$ Thus, both diagonal sums are equal.

(c) Substituting $a = 15$:
Diagonal 1: $15 + 23 = 38$
Diagonal 2: $16 + 22 = 38$
Hence, both are equal.

(d) The right number is always +1, the bottom is +7, and the diagonal is +8 from a. The total increase on both diagonals is always 8 added to 2 times a, regardless of the starting number.
One diagonal includes a and $a+8$, the other includes $a+1$ and $a+7$. In both cases, the total sum becomes:

$$2a + 8$$

Hence, for any value of a, the diagonals will sum to the same expression.

Answer: $2a+8, 38,$ $\begin{array}{|c|c|}\hline a & a+1 \\ \hline a+7 & a+8 \\ \hline \end{array}$

Keywords: $a+8, a+1+a+7, 2a+8, 38$

187. A triangle pattern is made using matchsticks as follows: Step 1 has 1 triangle, Step 2 has 2 joined triangles, and so on. The number of matchsticks used increases by 2 for each additional triangle.

(a) **Write the number of matchsticks for Step 1 through Step 5. What pattern do you observe?**

(b) **Find an algebraic expression to find the number of matchsticks at Step y.**

(c) **Use both expressions $3+2(y-1)$ and $2y+1$ to compute Step 25. Are they equal?**

(d) **Justify algebraically that $3+2(y-1)$ and $2y+1$ are equivalent expressions.**

Solution:

(a) The number of matchsticks used in each step is shown below:

Step (y)	Matchsticks
1	3
2	5
3	7
4	9
5	11

Pattern: Matchsticks increase by 2 for each step.

(b) Pattern follows: Matchsticks $= 3+2(y-1)$
Simplifying: $3+2y-2 = 2y+1$

(c) Using first expression:
$3+2(25-1) = 3+48 = 51$
Using second expression:
$2\times 25+1 = 50+1 = 51$
Both give same result.

(d) Simplify $3+2(y-1)$:

$$3+2y-2 = 2y+1$$

Thus, both expressions are algebraically equal.

Answer:

(a) Step 1–5: $3,\ 5,\ 7,\ 9,\ 11$
(b) Expressions: $3 + 2(y - 1),\ 2y + 1$
(c) Step 25: 51
(d) Justification: $3 + 2(y - 1) = 2y + 1$

Keywords: $3, 2y + 1, 51, 3 + 2(y - 1)$

188. Simplify the expression: $3a + 9b - 6 + 8a - 4b - 7a + 16$

(a) **Group and combine like terms from the expression.**
(b) **Rearrange using the commutative and associative properties of addition.**
(c) **Evaluate the simplified expression for** $a = 2, b = 1$**.**
(d) **Explain how the distributive and associative laws were used in simplification.**

Solution:

(a) Original expression: $3a + 9b - 6 + 8a - 4b - 7a + 16$
Group like terms:

$$(3a + 8a - 7a) + (9b - 4b) + (-6 + 16)$$

(b) Rearranging using properties:

$$(3a + 8a - 7a) = 4a, \quad (9b - 4b) = 5b, \quad (-6 + 16) = 10$$

Simplified expression:

$$4a + 5b + 10$$

(c) Substitute $a = 2, b = 1$:

$$4(2) + 5(1) + 10 = 8 + 5 + 10 = 23$$

(d) Distributive Law: Used when combining like terms, e.g., $a+a = 2a$, similar terms were factored mentally.
Associative Law: Used to group terms: $(3a + 8a - 7a)$, which simplified without changing the value.

Answer:

(a) Grouped: $3a + 8a - 7a,\ 9b - 4b,\ -6 + 16$
(b) Simplified: $4a + 5b + 10$
(c) Value: 23 for $a = 2,\ b = 1$
(d) Laws: Used distributive + associative to simplify structure

Keywords: $4a, 5b, 10, 4a + 5b + 10, 23$

189. A shop rents out chairs and tables. The rental rate is ₹40 per chair and ₹75 per table. Upon return, ₹6 is refunded per chair and ₹10 per table.

(a) **Write an expression for the total amount paid initially for renting x chairs and y tables.**

(b) **Write an expression for the total refund received on return.**

(c) **Write and simplify an expression for the final amount paid.**

(d) **Calculate the final amount if 5 chairs and 3 tables are rented.**

Solution:

(a) Initial amount paid:

$$\text{Chairs: } 40x, \quad \text{Tables: } 75y \quad \Rightarrow \quad \text{Total initial: } 40x + 75y$$

(b) Refund:

$$\text{Chairs: } 6x, \quad \text{Tables: } 10y \quad \Rightarrow \quad \text{Total refund: } 6x + 10y$$

(c) Final amount paid:

$$(40x + 75y) - (6x + 10y) = (40x - 6x) + (75y - 10y) = 34x + 65y$$

(d) Substitute $x = 5,\ y = 3$:

$$34(5) + 65(3) = 170 + 195 = 365$$

Answer:

(a) Initial: $40x + 75y$
(b) Refund: $6x + 10y$
(c) Final simplified: $34x + 65y$
(d) Final amount for $x = 5,\ y = 3$: ₹365

Keywords: $40x + 75y, 6x + 10y, 34x + 65y, 365$

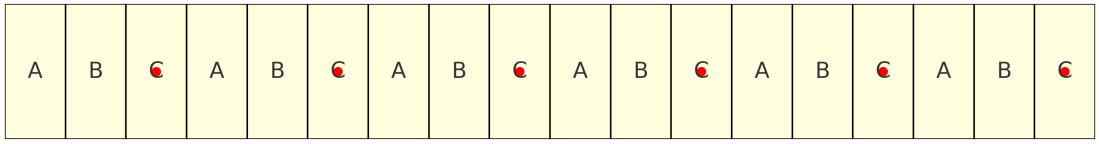
190. In a repeating border design, the sequence A, B, C repeats continuously as shown below.

(a) **What is the position of the 1st, 2nd, and 3rd occurrences of each design? Complete the table.**

(b) **Find algebraic expressions to describe the position of the nth appearance of designs A, B, and C.**

(c) **At which design will the 50th position fall? Justify using algebraic reasoning.**

(d) **Explain how the expressions relate to division by 3.**



Solution:

(a)

Occurrence	A	B	C
1st	1	2	3
2nd	4	5	6
3rd	7	8	9

(b) C appears every 3 steps: $\boxed{3n}$
B appears one before C: $\boxed{3n-1}$
A appears two before C: $\boxed{3n-2}$

(c) Divide 50 by 3: Quotient = 16, Remainder = 2
So, 50 is of the form $3n-1 \rightarrow$It is design **B**

(d) If remainder is:

- 0 $\rightarrow$position is multiple of 3 $\rightarrow$C
- 1 $\rightarrow$one after multiple of 3 $\rightarrow$A
- 2 $\rightarrow$one before multiple of 3 $\rightarrow$B

This is derived from the expressions in (b) using mod 3 logic.

Answer:

(a) 1st–3rd positions: A: 1,4,7; B: 2,5,8; C: 3,6,9
(b) Expressions: A: $3n-2,\ B: 3n-1,\ C: 3n$
(c) 50th position: Design B
(d) Remainders mod 3: 0, 1, 2

Keywords: $3n, 3n-1, 3n-2, 50 \div 3 = 16$ rem 2, Design B

191. Two expressions are given: $10y-3$ and $10(y-3)$. Answer the following:

(a) **Evaluate both expressions for $y = 0,\ 2,\ 5$.**
(b) **Simplify both expressions algebraically. Are they equal?**
(c) **Explain how the use of brackets affects the meaning and output.**
(d) **State whether the expressions are equivalent and justify your answer.**

Solution:

(a)

y	$10y - 3$	$10(y - 3)$
0	$0 - 3 = -3$	$10(-3) = -30$
2	$20 - 3 = 17$	$10(-1) = -10$
5	$50 - 3 = 47$	$10(2) = 20$

(b) Simplify both expressions:

$$10y - 3 \quad \text{(already simplified)}$$

$$10(y - 3) = 10y - 30$$

These are clearly not equal: $10y - 3 \neq 10y - 30$

(c) Brackets indicate that subtraction happens before multiplication. So: - $10y - 3$: Multiply first, then subtract 3. - $10(y - 3)$: Subtract first, then multiply by 10.

(d) They are not equivalent because their simplified forms differ and produce different values for the same input.

Answer:

(a) Values: $y = 0 \Rightarrow -3,\ -30;\ y = 2 \Rightarrow 17,\ -10;\ y = 5 \Rightarrow 47,\ 20$
(b) Simplified: $10y - 3$ and $10y - 30$
(c) Brackets change order of operations
(d) Not equivalent

Keywords: $10y - 3,\ 10(y - 3),\ -3,\ -30,\ 47,\ 20,$ Not equal

192. In a calendar-like number grid, the following cross-shaped pattern is observed. The center number is denoted by a.

(a) **Express all 5 numbers in the pattern in terms of a.**

(b) **Find the sum of these 5 numbers algebraically.**

(c) **Verify your result by substituting $a = 15$.**

(d) **Explain why the total is always 5 times the center number.**

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24	25	26	27	28	29	30
31	32	33	34	35	36	37
38	39	40	41	42	43	44
⋮	⋮	⋮	⋮	⋮	⋮	⋮

	?	
?	a	?
	?	

Solution:

(a)

	$a-7$	
$a-1$	a	$a+1$
	$a+7$	

(b) Sum:

$$(a-7)+(a-1)+a+(a+1)+(a+7)=5a$$

(c) Substituting $a=15$:
$(8+14+15+16+22)=75$
Also, $5\times 15=75$

(d) The outer numbers are symmetrically placed around the center. Their values increase and decrease in a balanced way:

$$(-7)+(-1)+0+1+7=0 \Rightarrow \text{sum becomes } 5a$$

So the pattern always totals $5a$.

Answer:

(a) $a-7,\ a-1,\ a,\ a+1,\ a+7$
(b) Sum: $5a$
(c) For $a=15,$ sum $=75$
(d) Symmetric cancellation $\rightarrow$total is always $5a$

Keywords: $a-7, a-1, a+1, a+7, 5a, 75$

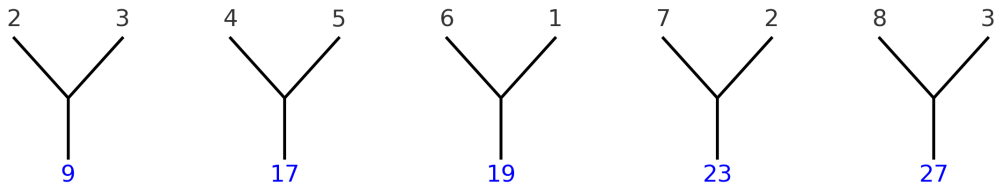
193. The diagram below shows a set of number machines. Each machine takes two inputs at the top: left input is a, right input is b. The machine follows a fixed rule to compute the output.

(a) **Identify the rule applied to generate the output. Write it as an algebraic expression.**

(b) **Use the expression to calculate the output when $a = 4$ and $b = 7$.**

(c) **Confirm that your rule works using two input-output pairs from the image.**

(d) **If the output is 30 and $b = 3$, find the value of a.**



Solution:

(a) From the image, observe:

(2, 3) $\rightarrow$3(2) + 3 = 9
(4, 5) $\rightarrow$3(4) + 5 = 17
(6, 1) $\rightarrow$3(6) + 1 = 19
The pattern is: Multiply the left input by 3, then add the right input.
The expression is: $\boxed{3a + b}$

(b) Substitute $a = 4,\ b = 7$:

$$3a + b = 3 \times 4 + 7 = 12 + 7 = 19$$

(c) Input $a = 7,\ b = 2$: $3 \times 7 + 2 = 21 + 2 = 23$
Input $a = 8,\ b = 3$: $3 \times 8 + 3 = 24 + 3 = 27$
These match the outputs shown in the diagram.

(d) Given $3a + b = 30$ and $b = 3$, we solve:

$$3a + 3 = 30 \Rightarrow 3a = 27 \Rightarrow a = 9$$

Answer:

(a) Expression: $3a + b$
(b) Output for $a = 4,\ b = 7:\ 19$
(c) Verified outputs: $(7, 2) = 23,\ (8, 3) = 27$
(d) a = 9 when output = 30 and b = 3

Keywords: $3a + b, 19, 23, 27, 9$

194. A row of squares is formed using matchsticks. Each square shares a side with the next.

(a) **Complete the table showing the number of matchsticks for 1 to 5 squares.**

(b) **Find an expression for the number of matchsticks needed to form s joined squares.**

(c) **How many matchsticks are needed to form 20 joined squares?**

(d) **Justify why your expression is valid using pattern reasoning.**

Solution:

(a)

No. of Squares (s)	Matchsticks Used
1	4
2	7
3	10
4	13
5	16

(b) Each additional square adds 3 more matchsticks (since one side is shared).
Pattern: Matchsticks $= 4 + 3(s - 1)$
Simplifies to: $\boxed{3s + 1}$

(c) Substitute $s = 20$:

$$3 \times 20 + 1 = 60 + 1 = 61$$

(d) The first square requires 4 matchsticks. Each additional square shares one side, adding only 3 matchsticks.
So for s squares, we have:

$$4 + 3(s - 1) = 3s + 1$$

This matches the observed pattern and simplifies the count.

Answer:

(a) Values: $4,\ 7,\ 10,\ 13,\ 16$
(b) Expression: $3s + 1$
(c) For $s = 20,$ matchsticks $= 61$
(d) Justification: 1st square = 4, others add 3 each

Keywords: $4, 3s + 1, 61,$ shared side, tiling squares

195. In a quiz competition, Charu scores marks for correct answers and loses marks for wrong answers. Her scores for the three rounds are given as:

$$\text{Round 1: } 6p-2q, \quad \text{Round 2: } 8p-3q, \quad \text{Round 3: } 7p-4q$$

(a) **Find the expression for Charu's total score over all three rounds. Simplify the result.**

(b) **Find her final score if $p=4$ and $q=1$.**

(c) **Krishita's total score is given as $21p-7q$. Who scored more?**

(d) **How many more marks did the higher scorer get? Express as a simplified expression.**

Solution:

(a) Total score:

$$(6p-2q)+(8p-3q)+(7p-4q)=(6+8+7)p-(2+3+4)q=21p-9q$$

(b) Substitute $p=4,\ q=1$:

$$21\times 4-9\times 1=84-9=75$$

(c) Krishita's score:

$$21p-7q \text{ at } p=4,\ q=1=84-7=77$$

So Krishita scored more.

(d) Difference:

$$(21p-7q)-(21p-9q)=-7q+9q=2q$$

Substitute $q=1$: $2\times 1=2$

Answer:

(a) Total: $21p-9q$
(b) Charu's score: 75
(c) Krishita's score: 77 (higher)
(d) Difference: $2q=2$

Keywords: $21p-9q, 75, 77, 2q$, Krishita scored more

196. A rope is folded and then cut once. Each fold creates one extra layer, so each fold increases the number of resulting pieces by 1.

(a) **Complete the table for number of folds vs number of resulting rope pieces.**

(b) **Write an expression to calculate the number of pieces after r folds.**

(c) **How many pieces will result from 15 folds and one cut?**

(d) **Justify why your expression is valid for all values of $r \geq 0$.**

Solution:

(a)

No. of Folds (r)	No. of Pieces
0	2
1	3
2	4
3	5
4	6

(b) The number of pieces increases by 1 with each fold, starting from 2 when no folds are made:

$$\text{Expression: } \boxed{r + 2}$$

(c) For $r = 15$:

$$15 + 2 = 17 \text{ pieces}$$

(d) Each fold doubles a section of the rope over another. With no folds, a single cut gives 2 parts. Each additional fold adds one more layer to cut through, resulting in one additional piece. So for any number of folds r, the formula $r + 2$ holds.

Answer:

(a) Table: $2,\ 3,\ 4,\ 5,\ 6$
(b) Expression: $r + 2$
(c) For $r = 15,$ pieces $= 17$
(d) Justification: Each fold adds one new piece

Keywords: $2,\ 3,\ 4,\ r + 2,\ 17$

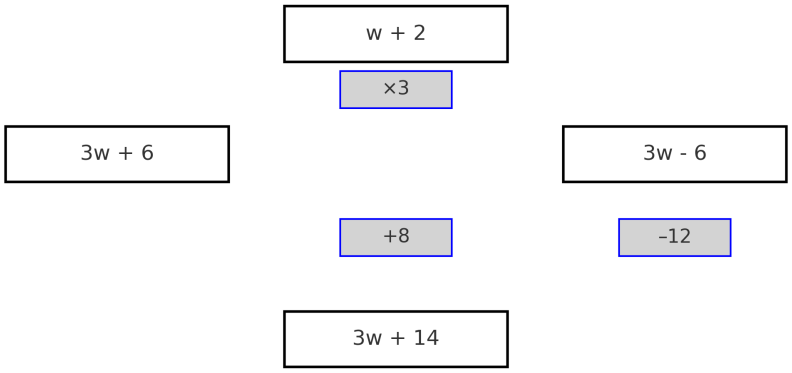
197. The diagram below shows a puzzle where the expression $w + 2$ moves through various operations along three different paths. Use the diagram to answer the following:

(a) **Starting from $w + 2$, find the simplified expression on the left path.**

(b) **Which operations lead to the expression $3w + 14$? Describe the sequence.**

(c) **What value of w gives the same result for both expressions: $3w + 6$ and $3w - 6$? Justify.**

(d) **If the final output is 18 on the center path, what is the value of w?**



Solution:

(a) Left path: Multiply $w + 2$ by 3:

$$3(w + 2) = 3w + 6$$

(b) Center path: First multiply $w + 2$ by 3 $\rightarrow 3w + 6$, then add 8:

$$3w + 6 + 8 = 3w + 14$$

(c) Set $3w + 6 = 3w - 6 \Rightarrow 3w + 6 - 3w = -6 \Rightarrow 6 = -6$ — contradiction. So, no such value of w exists where both expressions are equal.

(d) On center path: $3w + 14 = 18 \Rightarrow 3w = 4 \Rightarrow w = \frac{4}{3}$

Answer:

(a) $3w + 6$ (b) Path: $\times 3$, then $+8$
(c) No such w exists where $3w + 6 = 3w - 6$
(d) $w = \dfrac{4}{3}$

Keywords: $3w + 6,\ 3w + 14,\ 3w - 6,\ w = \frac{4}{3}$

198. Simplify the following algebraic expressions step-by-step. Answer each subpart carefully.

(a) **Simplify:** $4a + 3b - 2a + 5 - b + 8$

(b) **Simplify using distributive law:** $3(p+4q-2)-2(p-q+1)$

(c) **Simplify and combine like terms:** $2x-[3x-2(x+5)]+4$

(d) **Evaluate your final expression from (b) for** $p=2, q=-1$

Solution:

(a)

$$4a-2a+3b-b+5+8=2a+2b+13$$

(b)

$3(p+4q-2)=3p+12q-6-2(p-q+1)=-2p+2q-2$Combine: $(3p-2p)+(12q+2q)+(-6-2)=$

(c)

$$2x-[3x-2(x+5)]+4=2x-[3x-2x-10]+4=2x-[x-10]+4=2x-x+10+4=x+14$$

(d) From (b), expression was $p+14q-8$. Substituting $p=2, q=-1$:

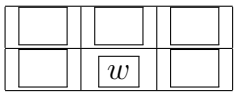
$$2+14(-1)-8=2-14-8=-20$$

Answer:

(a) $2a+2b+13$
(b) $p+14q-8$
(c) $x+14$
(d) -20

Keywords: $2a+2b+13,\ p+14q-8,\ x+14,\ -20$

199. A 2 × 3 grid of numbers is shown below. The middle bottom cell is labeled w. Use the pattern of increasing numbers to fill in the remaining cells.



(a) **Write expressions for all five other blank cells in terms of** w**.**

(b) **If** $w=23$**, fill in all the cells using your expressions.**

(c) **What is the total of all six numbers in the grid in terms of** w**?**

(d) **Verify the total when** $w=23$**.**

Solution:

(a) Let the grid look like this:

$w-8$	$w-7$	$w-6$
$w-1$	w	$w+1$

(b) If $w = 23$:

15	16	17
22	23	24

(c) Sum = $(w-8)+(w-7)+(w-6)+(w-1)+w+(w+1)$
Combine like terms:

$$6w - 21$$

(d) Substitute $w = 23$: $6 \times 23 - 21 = 138 - 21 = 117$

Answer:

(a) $w-8,\ w-7,\ w-6,\ w-1,\ w+1$
(b) Values: $15,\ 16,\ 17,\ 22,\ 24$
(c) Total: $6w - 21$
(d) If $w = 23,$ total = 117

Keywords: $w-8, w+1, 6w-21, 117$

200. In a grid puzzle, each row follows a different algebraic rule. The first value in each row is given. Complete each row based on the rule.

Row	Box 1	Box 2	Box 3
1	2		
2	$a = 5$		
3	$b = 7$		

(a) **Row 1: Rule →Multiply by 3 and subtract 1. Fill all boxes.**
(b) **Row 2: Rule →Multiply the previous number by 2. Let $a = 5$. Fill all boxes.**
(c) **Row 3: Rule →Add 4 to the previous number. Let $b = 7$. Fill all boxes.**
(d) **Write algebraic expressions for Box 2 and Box 3 of each row in terms of the first box.**

Solution:

(a) **Given:** First number = 2, rule = ×3 then −1
To calculate: Box 2 = $3 \times 2 - 1 = 5$
Box 3 = $3 \times 5 - 1 = 14$
Final values: Row 1 →2, 5, 14

(b) **Given:** First number $a = 5$, rule = multiply by 2
To calculate: Box 2 = $2 \times 5 = 10$
Box 3 = $2 \times 10 = 20$
Final values: Row 2 $\rightarrow$5, 10, 20

(c) **Given:** First number $b = 7$, rule = add 4 to previous
To calculate: Box 2 = $7 + 4 = 11$
Box 3 = $11 + 4 = 15$
Final values: Row 3 $\rightarrow$7, 11, 15

(d) **Row 1:**

Box 2: $3x - 1,$ Box 3: $3(3x - 1) - 1 = 9x - 4$

Row 2:

Box 2: $2a,$ Box 3: $2 \times 2a = 4a$

Row 3:

Box 2: $b + 4,$ Box 3: $b + 8$

Answer:

(a) Row 1: $2,\ 5,\ 14$
(b) Row 2: $5,\ 10,\ 20$
(c) Row 3: $7,\ 11,\ 15$
(d) Expressions: Row 1: $3x - 1,\ 9x - 4$; Row 2: $2a,\ 4a$; Row 3: $b + 4,\ b + 8$

Keywords: $3x - 1,\ 9x - 4,\ 2a,\ 4a,\ b + 4,\ b + 8,\ 2, 5, 14$